Raab Game II

Time limit: 1.00 s **Memory limit:** 512 MB

Consider a two player game where each player has n cards numbered $1, 2, \ldots, n$. On each turn both players place one of their cards on the table. The player who placed the higher card gets one point. If the cards are equal, neither player gets a point. The game continues until all cards have been played.

You are given the number of cards n and the players' scores at the end of the game, a and b. Your task is to count the number of possible games with that outcome.

Input

The first line contains one integer t: the number of tests.

Then there are t lines, each with three integers n, a and b.

Output

For each test case print the number of possible games modulo $10^9 + 7$.

Constraints

- $1 \le t \le 1000$
- $1 \le n \le 5000$
- $0 \le a, b \le n$

Example

Input	Output
5 3 1 2 2 0 1 5 2 2 9 3 5 4 4 1	6 0 4200 976757050 0